Practice Exercises for Module 7.

Problem 1 (Validation Set): Consider the following dataset with $N = 6$ and $p = 1$:

$$\mathcal{D} = \{(1, 1.2), (2, 1.8), (3, 3.5), (4, 4.7), (5, 4.7), (6, 5.4)\}$$

We would like to compare the performance of the following three models (feel free to use software to learn the parameters of each model; there is no need to do the computations by hand) which ones of the models would you use in practice?

$$\mathcal{M}_0 : \widehat{Y}_0 = \widehat{\alpha}_0$$
$$\mathcal{M}_1 : \widehat{Y}_1 = \widehat{\beta}_0 + \widehat{\beta}_1 X$$
$$\mathcal{M}_2 : \widehat{Y}_2 = \gamma_0 + \gamma_1 X + \gamma_2 X^2$$

a) Use the validation set approach to estimate the testing error of the three models above assuming that the data set was randomly split into the following training and validation subsets :

$$\mathcal{D}_{Train} = \{(1, 1.2), (3, 3.5), (4, 4.7)\}$$
$$\mathcal{D}_{Test} = \{(2, 1.8), (5, 4.7), (6, 5.4)\}$$

b) Compare the training and the testing errors for the three models above. Is the Training Error decreasing with the flexibility of your model?

c) Based on this result, which ones of the models would you use in practice?

Problem 2 (k-Fold Cross-Validation): Consider the following dataset with $N = 9$ and $p = 1$:

$$\mathcal{D} = \{(1, 1.2), (2, 1.8), (3, 3.5), (4, 4.7), (5, 4.7), (6, 5.4), (7, 7.1), (8, 7.9), (9, 8.1)\}$$

We would like to compare the performance of the following three models (feel free to use software to learn the parameters of each model; there is no need to do the computations by hand):

$$\mathcal{M}_0 : \widehat{Y}_0 = \widehat{\alpha}_0$$
$$\mathcal{M}_1 : \widehat{Y}_1 = \widehat{\beta}_0 + \widehat{\beta}_1 X$$
$$\mathcal{M}_1 : \widehat{Y}_1 = \gamma_0 + \gamma_1 X + \gamma_2 X^2$$

a) Use the 3-fold cross-validation to estimate the testing error of the three models above assuming that the data set was randomly split into the following three subsets (feel free to use software to train your models):

$$\mathcal{D}_1 = \{(3, 3.5), (5, 4.7), (8, 7.9)\}$$
$$\mathcal{D}_2 = \{(1, 1.2), (4, 4.7), (7, 7.1)\}$$
$$\mathcal{D}_3 = \{(2, 1.8), (6, 5.4), (9, 8.1)\}$$

b) Is the testing Error decreasing with the flexibility of your model?

c) Based on this result, which ones of the models would you use in practice?

Problem 3 (Bootstrap): Consider the following dataset with $N = 3$ and $p = 1$:

$$\mathcal{D} = \{(1, 1.2), (2, 1.8), (3, 3.5), (4, 4.7)\}$$

We would like to estimate the uncertainty in the parameters of the following linear model: $\widehat{Y}_1 = \widehat{\beta}_0 + \widehat{\beta}_1 X$. To achieve this, we will use the bootstrap method considering that the following datasets are three bootstrap samples (each dataset is obtained by sampling four random points from $\mathcal{D}$, with replacement):

$$\mathcal{D}_1 = \{(1, 1.2), (3, 3.5), (1, 1.2), (4, 4.7)\}$$
$$\mathcal{D}_2 = \{(4, 4.7), (3, 3.5), (2, 1.8), (4, 4.7)\}$$
$$\mathcal{D}_3 = \{(1, 1.2), (4, 4.7), (1, 1.2), (4, 4.7)\}$$

(As in previous problems, feel free to use software to learn the parameters of each model; there is no need to do the computations by hand)

a) Use the bootstrap technique to estimate the standard deviation of the intercept $\widehat{\beta}_0$.

b) Use the bootstrap technique to estimate the standard deviation of the slope $\widehat{\beta}_1$.